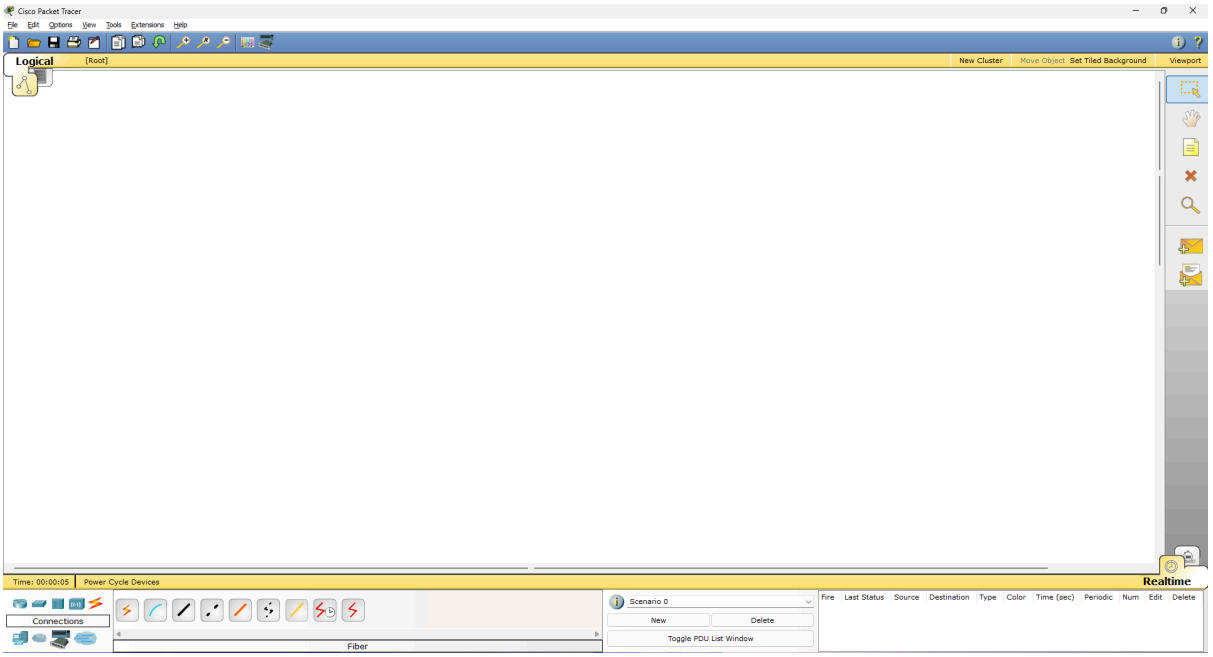
PRACTICALS

Computer Networks Laboratory

Practical 9.

Aim: Explore the utility of Virtual LANs (VLANs) by constructing a network infrastructure configured with multiple distinct VLANs to manage broadcast domains and segment traffic.

Step 1: Open Cisco packet tracer 5.1 as shown in **FIG 1**

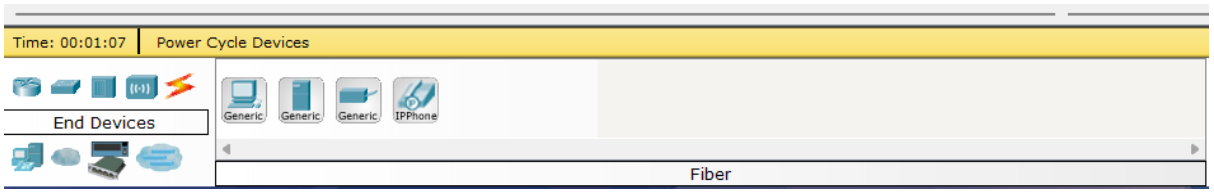


[FIG 1]

Step 2: First we will Understand how to build Topology then we will create various topologies. Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection



[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 3 - PCs
2. 1 - Network Switch
3. 3 - Connecting wires/ cables



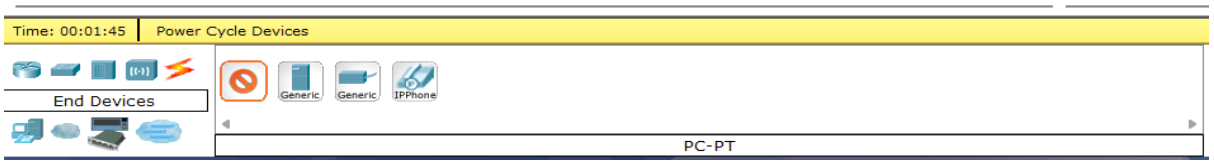


FIG 3

Step 4: Creating Network (Topology) Using selected devices as shown in **FIG 4**

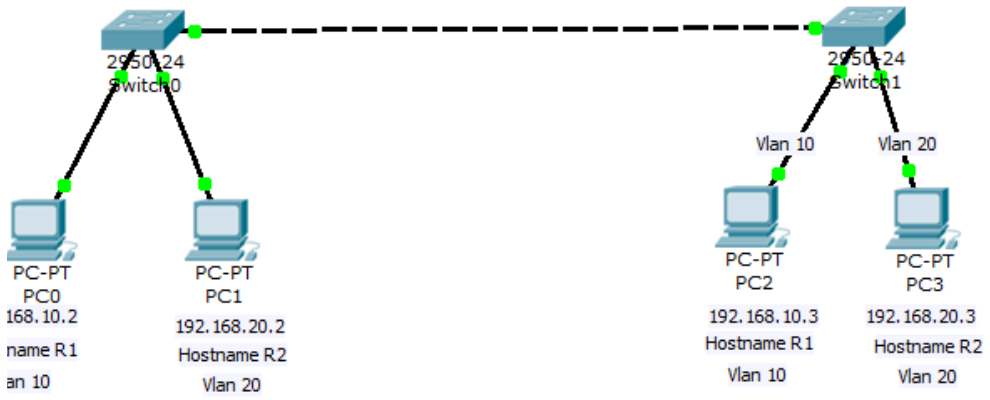


FIG 4

Step 5: Configuring The Setup with Network (static method) Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration

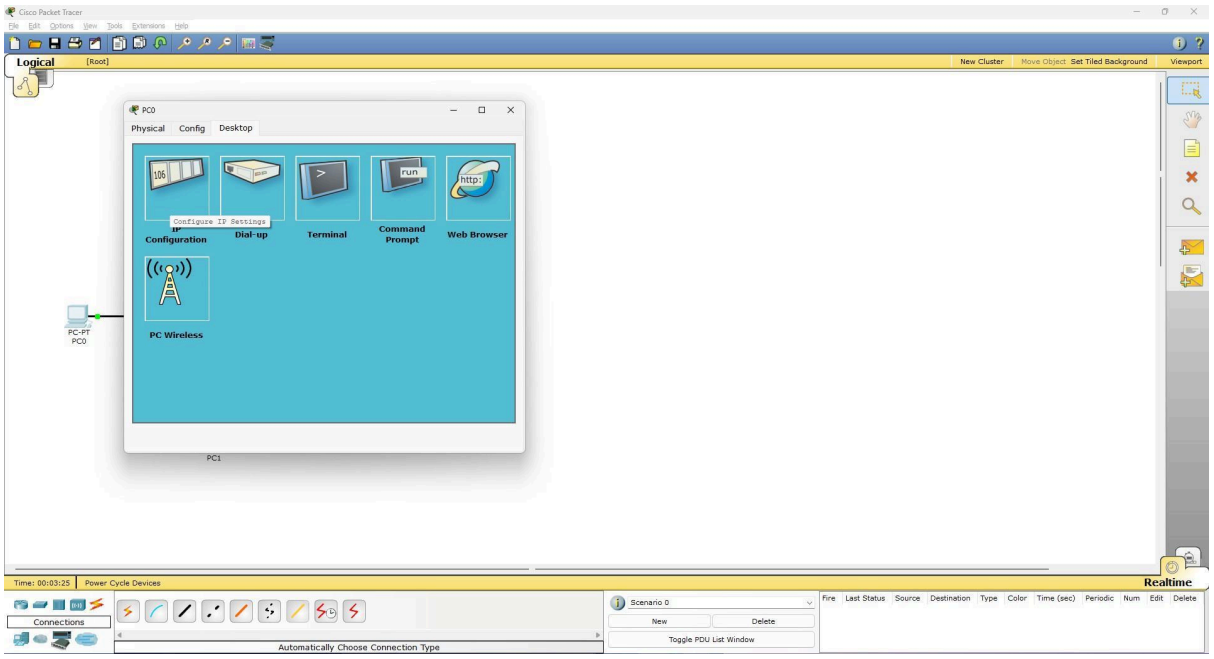


FIG 5

1. Select Or click on Static
2. Enter values as follows
3. Close the dialog box
4. Do the same process for all of the PCs

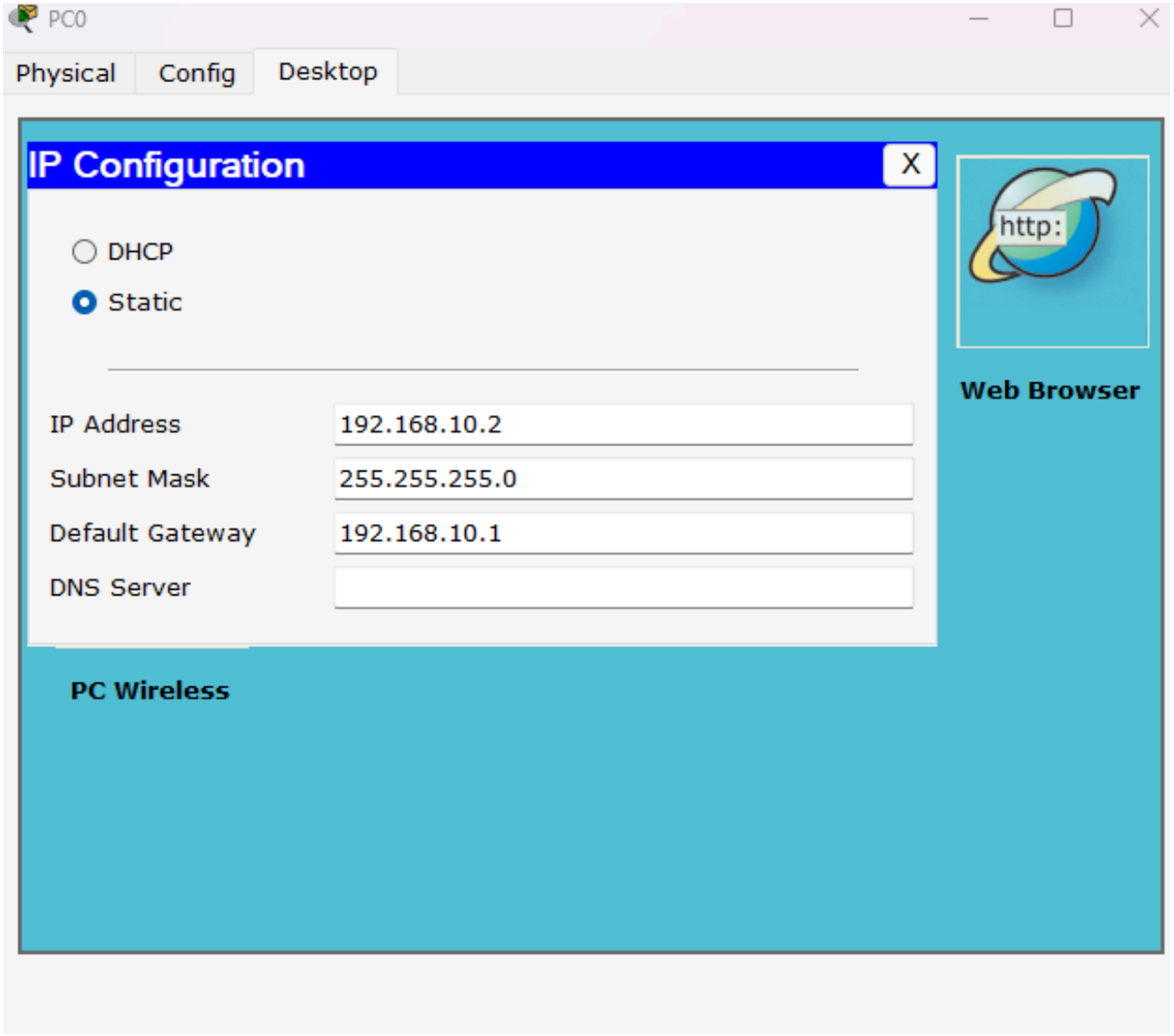


FIG 6

Step 6: Now Click on Switch and Go to CLI and write the Following Command

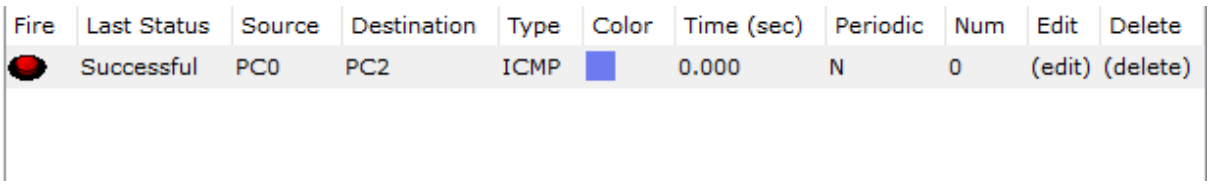
Here we are going to setup 2 Vlans in Switch 0 (Vlan 10 for Pc 0, PC 2 and Vlan 20 for Pc 1, PC 3)

```
enable
configure terminal
hostname R1
vlan 10
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit
hostname R2
vlan 20
interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
exit
interface fastEthernet 0/24
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
exit
```

Here we are going to setup 2 Vlans in Switch 1 (Vlan 10 for Pc 0, PC 2 and Vlan 20 for Pc 1, PC 3)

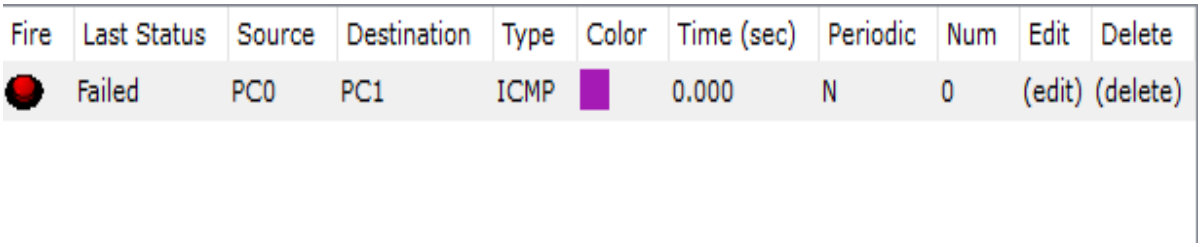
```
enable
configure terminal
hostname R1
vlan 10
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit
hostname R2
vlan 20
interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
exit
interface fastEthernet 0/24
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
exit
```

Step 7: After compilation of step 6 in the bottom Tool bar's right side we can check the status of sending packets that it's successfully Send or Failed Shown in **FIG 8.1** For Success and **FIG 8.2** For Failed



Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)

FIG 8.1



Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)

FIG 8.2

Make Sure if you share Packets from one Pc to another then it show be in same Vlan then only it will share packet else it will fail to share packet as shown in **FIG 8.2** Here in this Topology PC 0 and PC 2 are in Vlan 2 and PC 1 and PC 3 in another Vlan so PC can share with PC 2 but can't with PC1 Or PC 3